

NAVISENCE

Environment Detection for Blind People

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Project Advisor

Prof. Muhammad Fahad

Project Manager

Dr. Muhammad Ilyas

Project Team

Muhammad Ali Raza Ansari (BSCS51F22S008) — Team Leader

Muhammad Abdullah Zammad (BSCS51F22S036) — Team Member

Muhammad Hassan Javed (BSCS51F22S040) — Team Member

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Project Supervisor's Signature

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1. Abstract

Visually impaired individuals face significant challenges in perceiving their surroundings in both indoor and outdoor environments, where objects are constantly changing and moving. Navisence is a smartphone-based assistive system that uses real-time object detection and audio feedback to help visually impaired users navigate safely and independently. The system captures live video through the smartphone camera, detects surrounding objects in both indoor and outdoor settings, and communicates results to the user via spoken alerts through headphones or a speaker, along with left, right, and center directional guidance and near, far, and medium distance estimation. Navisence operates entirely offline without requiring any internet connection, ensuring reliable and uninterrupted assistance in all environments. Using only built-in smartphone hardware, Navisence provides a simple, portable, and cost-effective solution that reduces accidents and promotes independence in everyday environments.

2. Background and Justification

According to the World Health Organization (WHO) [1], approximately 2.2 billion people worldwide experience vision impairment, emphasizing the need for effective assistive technologies. Traditional aids such as white canes and guide dogs help detect nearby obstacles but offer limited environmental awareness and cannot identify surrounding objects [2].

Several Electronic Travel Aids (ETAs), including SmartCane, Ultra Cane, and WeWALK, use ultrasonic sensors and haptic feedback to assist users [3][4]. Additionally, AI-based wearable devices such as OrCam MyEye and Envision Glasses provide real-time auditory feedback using camera-based systems [5][6]. While these solutions improve mobility, ultrasonic devices cannot distinguish object types, and AI-powered wearables are often expensive, internet-dependent, and inaccessible in areas with poor or no connectivity.

Therefore, there is a need for a fully offline, low-cost, portable, and user-friendly solution that utilizes a smartphone camera to detect surrounding objects and provide real-time audio feedback without requiring any internet connection. The proposed system, Navisence, aims to address this gap by offering an affordable, offline-capable assistive tool to enhance environmental awareness for visually impaired individuals in any setting.

3. Project Methodology

Phase 1: Requirement Analysis

System requirements are defined for a mobile-based assistive application, focusing on real-time image capture, object detection using computer vision, offline audio feedback through Text-to-Speech (TTS), and left, right, and center directional indication with near, far, and medium distance estimation. Performance requirements such as low latency, accuracy, offline functionality, and usability on smartphones are also identified.

Phase 2: Application Development

A mobile application is developed using Flutter to access the smartphone camera and capture real-time video frames. Flutter's built-in camera handling capabilities are utilized for image acquisition and preprocessing to prepare frames for object detection. The application is designed to function fully offline without requiring any internet connection.

Phase 3: Model Integration

A TensorFlow Lite-based computer vision model is integrated into the Flutter application to detect objects from the captured frames. TensorFlow Lite is specifically chosen for its lightweight, on-device processing capability, allowing the model to run entirely offline in real time to efficiently identify surrounding objects in both indoor and outdoor environments.

Phase 4: Audio and Directional Feedback

Detected objects are converted into speech using an offline Text-to-Speech (TTS) engine. The system determines the relative position of each detected object within the camera frame and provides audio feedback indicating the direction (left or right) and approximate distance (near, medium, or far) of the object, all without requiring any internet connection.

Phase 5: Testing and Refinement

The application is tested across various indoor and outdoor environments, including areas with no internet connectivity, to evaluate detection accuracy, response time, and overall reliability. Based on the results, necessary refinements are made to improve offline system performance and user experience.

4. Project Scope**Included:**

- Real-time object detection using a smartphone
- Fully offline operation — no internet connection required
- Audio feedback using offline Text-to-Speech (TTS)
- Direction identification (Left, Right, Center)
- Distance estimation (Near, Far, Medium)
- Continuous real-time system operation
- Simple and accessible mobile user interface

Not Included:

- Exact distance estimation between the user and objects
- Object size measurement
- Navigation or path planning features
- IoT devices or external hardware integration, except for smartphones
- Facial recognition functionality
- Text reading or Optical Character Recognition (OCR)

5. High-Level Project Plan

CP1 — Project Plan

Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Phase 1: Project Proposal												
Phase 2: SRS Development												
Phase 3: Prototype Design												
Phase 4: Final Documentation												

CP2 — System Development Gantt Chart (12 Weeks)

Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Phase 1: Dataset Preparation												
Phase 2: Model Training												
Phase 3: App Setup & Model Integration												
Phase 4: Testing												

6. References

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